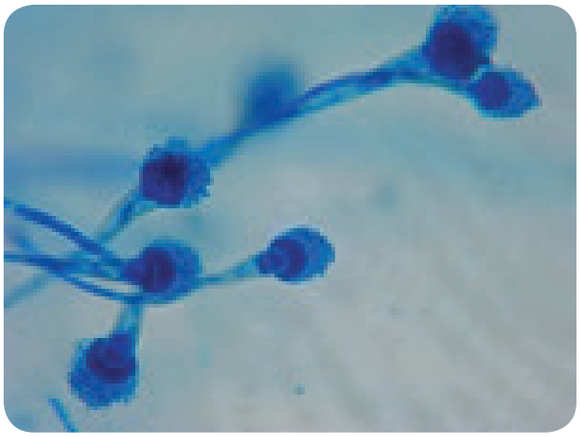


Figure 9.2: Microscopic view of yeast Moulds

Phaeoid fungi: Most of true pathogenic fungi are dimorphic fungi which are composed of darkly coloured hyphal form known as **dematiaceous fungi**. Some are yeast like and also known as black yeasts.

Vegetative Structures: Several structures are formed by the vegetative mycelia that have no reproductive value but are important for the differentiation of fungi eg. Chlamydospores and Arthrospores. Chlamydospores are thick walled, resistant to adverse conditions and are larger than other cells. Arthrospores are rectangular spores which are thick walled that are disposed on maturity.

ii. Cell structure

a. Capsule: Fungi produce an extra cellular polysaccharide in the form of capsule. Example: *Cryptococcus*.

b. Cell wall: Fungi possess a multilayered rigid cell wall exterior to the plasma lemma. The cell wall is made up of chitin, a water insoluble, homo polymer of N-acetyl glucosamine. Chitin synthase is responsible for the bio synthesis of chitin.

c. Plasma lemma: Cytoplasmic membrane or plasma lemma encloses complex cytosol. It is composed of glycoprotein, lipids and ergosterol.

d. Cytosol: Cytosol comprises of mitochondria, microtubules ribosome's, golgi apparatus, double membrane endoplasmic reticulum and Nucleus. The nuclei of the fungi are enclosed by a membrane and contain most of cellular DNA.

iii. Reproduction of fungi

Spores play a major role in reproduction. There may be asexual or sexual cell divisions.

a. Asexual Reproduction: The asexual reproduction involves, budding or fission or mitosis. Fungi produce more than one type of asexual spores. They are microspores (micro conidia) and macrospores (macro conidia). Spores that are present inside sporangium are

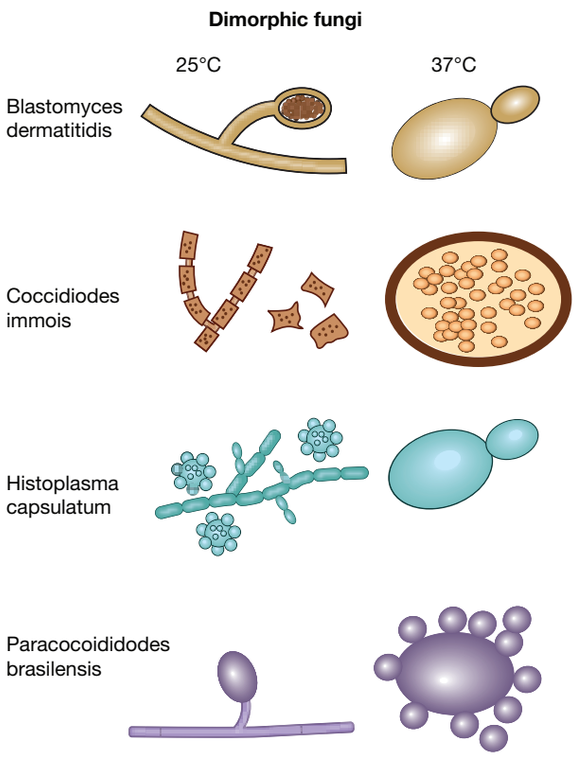


Figure 9.3: Dimorphic Fungi

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